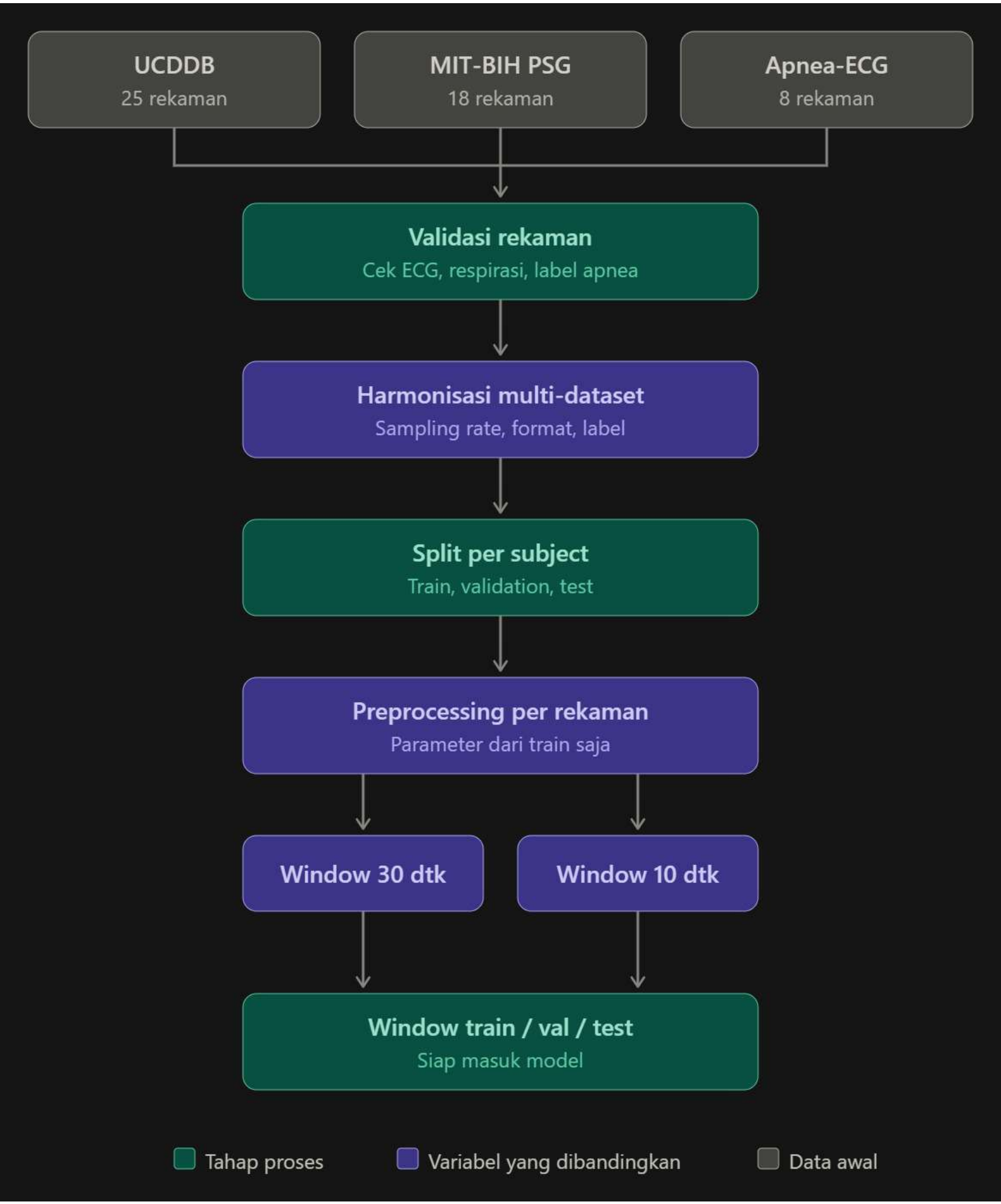
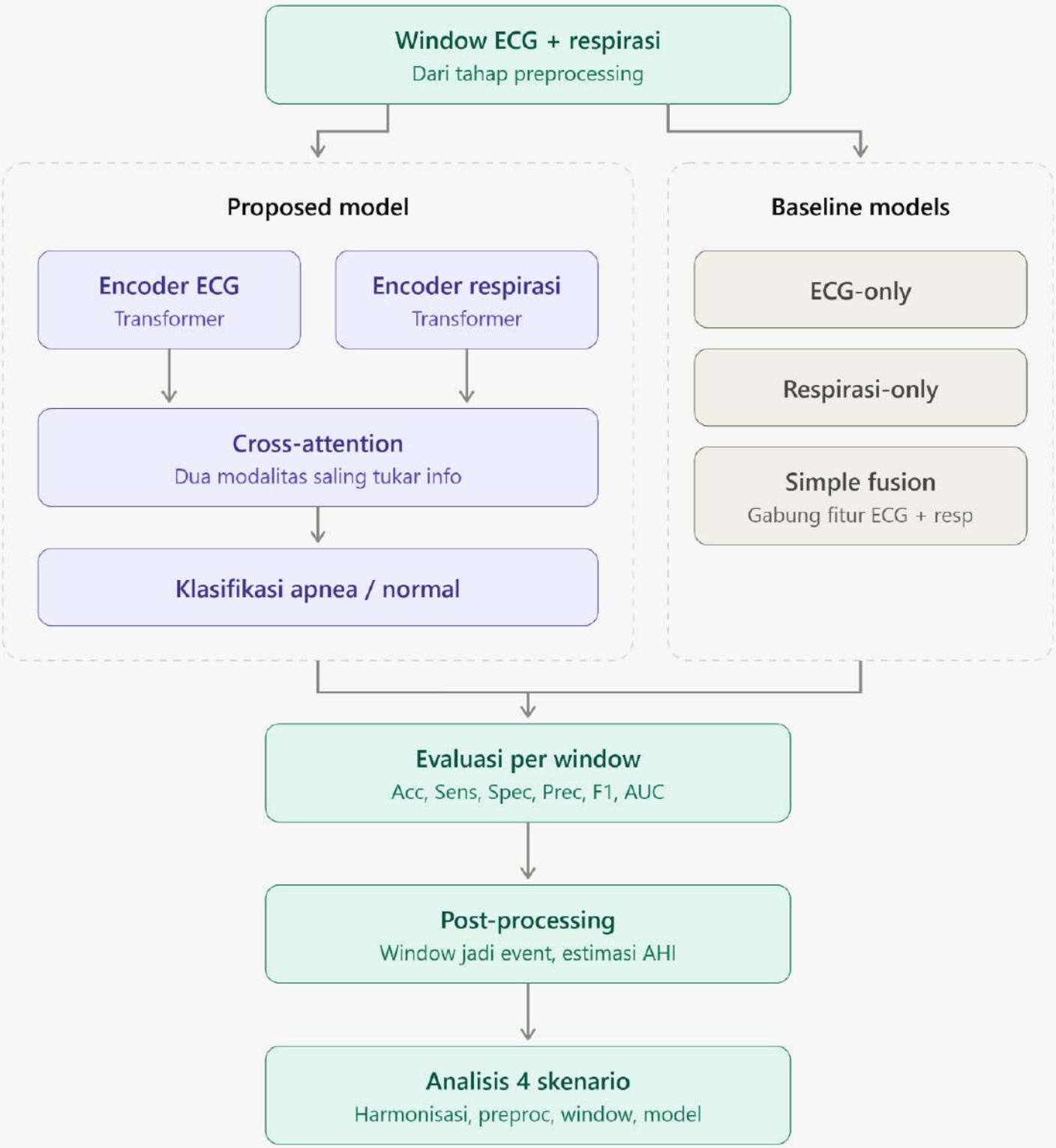

UCDDB
25 rekaman
MIT-BIH PSG
18 rekaman
Apnea-ECG
8 rekaman
Validasi rekaman
Cek ECG, respirasi, label apnea
Harmonisasi multi-dataset
Sampling rate, format, label
Split per subject
Train, validation, test
Preprocessing per rekaman
Parameter dari train saja
Window 30 dtk
Window 10 dtk
Window train / val / test
Siap masuk model
Tahap proses
Variabel yang dibandingkan
Data awal


Window ECG + respirasi
Dari tahap preprocessing
Proposed model
Encoder ECG
Transformer
Encoder respirasi
Transformer
Cross-attention
Dua modalitas saling tukar info
Klasifikasi apnea / normal
Baseline models
ECG-only
Respirasi-only
Simple fusion
Gabung fitur ECG + resp
Evaluasi per window
Acc, Sens, Spec, Prec, F1, AUC
Post-processing
Window jadi event, estimasi AHI
Analisis 4 skenario
Harmonisasi, preproc, window, model